

PRODUCTION ASSET REGISTER

Baseline Capital Infrastructure Ledger

Systematic Cataloging, Depreciation Lifecycles, & Engineering Parameters

Asset Categorization & Auditing Protocol

This register provides institutional investors with a completely clear, transparent record of all baseline equipment components purchased with the capital allocation.

Asset Class ID	Component Domain Focus	Depreciation Method	Verification Frequency
LWP-STR-001	Civil Works & Structural Shell Enclosures	Straight-Line (20 Years)	Bi-Annual Structural Audit
LWP-ENG-002	Off-Grid Bio-Solar Utility Infrastructure	Straight-Line (15 Years)	Quarterly Power Quality Test
LWP-HYD-003	Hydrological Lifelines & Biosecurity Rigs	Straight-Line (10 Years)	Monthly Pathogen-Barrier Audit

Structural Assets: Broiler Sprint Barn

Asset Code: LWP-STR-PLTY

- ✓ **Description:** 384 m² High-density open-sided commercial poultry barn built to house 5,760 birds.
- ✓ **Material Grade:** Industrial galvanized steel framing with protective anti-corrosion zinc coatings.
- ✓ **Civil Foundation:** 150mm reinforced concrete slab with perimeter deep footing lines.



Engineering Value

The structural layout is designed to handle highveld wind loads while optimizing passive zero-watt interior ventilation paths.

Structural Assets: Swine Finisher Barn

Asset Code: LWP-STR-SWN

- ✓ **Description:** 384 m² Open-sided grow-out barn structured into 16 discrete containment pens.
- ✓ **Biological Design:** Handles 160 market hogs comfortably at a generous 1.2 m² per head allocation.
- ✓ **Logistics Corridors:** Integrated 1m wide concrete central access paths for stress-free animal sorting.



Pennage Security

Features heavy-duty steel pipe wall dividers securely anchored directly into the floor slab to withstand animal movement stresses.

Isolation Assets: 4-Pod Nursery Block

Asset Reference	Component Breakdown	Material Standard	Critical Protection Metric
LWP-ISO-NURS	4-Pod Airtight Swine Nursery Block (48 m² Total Area)	Double-plastered masonry bricks with smooth antimicrobial surface coatings.	Creates completely isolated, decoupled viral rooms to restrict outbreaks to single groups.
LWP-ISO-WALL	150mm Internal Structural Firewalls	Solid masonry core blocks with absolute floor-to-ceiling seals.	Prevents any airborne cross-contamination or moisture leaks between adjacent nursery chambers.

Climate Control Assets: Curtains & Insulation

Asset Group: LWP-CLM-SHLD

- ✓ **PVC Curtain Arrays:** 550g/m² UV-stabilized heavy-duty PVC curtains paired with industrial manual winch pulley rigs.
- ✓ **Dwarf Wall Interface:** 600mm High plastered masonry wall base built into open-sided barn templates.
- ✓ **Thermal Shield Panels:** 50mm High-density polyisocyanurate (PIR) insulation backing plates fitted under the roof profiles.



Thermal Regulation

The dwarf walls stop floor-level freezing drafts during winter brooding, while the PIR insulation cuts active solar heat loads by up to 35%.

Energy Assets: Off-Grid 30kW/30kWh BESS

Asset Code: LWP-PWR-CORE

- ✓ **Inverter Sub-System:** 30 kW Smart industrial hybrid inverters featuring dual high-voltage MPPT inputs.
- ✓ **Storage Battery Bank:** 30 kWh Lithium Iron Phosphate (LFP) rack-mount cells with integrated smart BMS modules.
- ✓ **Primary Fuel Array:** High-efficiency monocrystalline solar PV modules clamped to custom barn roof mounts.



Utility Resilience

This asset bypasses all local power grid dependencies, running our production setups inside a protected ZAR 1.80/kWh baseline model cost zone.

Energy Assets: Biogas Digestion Loop

Asset Identifier	Component Technical Name	Operational Input	Co-Generation Utility Return
LWP-BIO-DIGS	Fixed-Dome Anaerobic Digestion Reactor Tank	Raw manure slurry from swine concrete sub-gutters.	Generates high-purity methane gas stored in low-pressure rubber storage bags.
LWP-BIO-GENS	Modified Methane GenSet Engine & Heat Exchanger Loop	Scrubbed biological methane fuel source.	Delivers supplementary night-time electrical volts and steam heat for nursery floors.

Utility Assets: Hydrological Pumping Rig

Asset Group: LWP-HYD-LINE

- ✓ **Submersible Pump Array:** High-head solar-powered DC submersible pump lowered into a verified borehole line.
- ✓ **Elevated Storage Towers:** Reinforced steel framing holding multi-tank heavy-duty JoJo arrays.
- ✓ **Operational Security Limit:** Holds an engineered 5-day on-site water supply safety buffer.



Hydraulic Stability

The system uses high towers to supply water via gravity alone, keeping all automatic drinker nipples pressurized even if the pumps are down for maintenance.

Sanitization Assets: Inline Purification Rig

Asset Identifier	Purification Component	Technical Engineering Target	Biological Security Impact
LWP-SAN-FILT	50-Micron Kinetic Disc Filtration Bank	Removes suspended physical sand and clay particles down to 50 microns.	Prevents fine sediment from blocking down-line automated drinker valves.
LWP-SAN-STER	Inline UV Ultraviolet Sterilization & Chlorination Rig	High-intensity UV exposure chambers linked to chemical dosing dispensers.	Neutralizes waterborne viral chains, keeping internal pipelines clear of biofilms.

Data Assets: LoRaWAN IoT Sensor Network

Asset Code: LWP-IOT-NERV

- ✓ **Central Gateway Node:** Outdoor industrial IP67-rated LoRaWAN internet gateway module.
- ✓ **Air Environment Probes:** Digital sensor nodes measuring real-time temperature, ambient humidity, and ammonia gas levels.
- ✓ **Liquid Transit Probes:** Clamp-on ultrasonic non-invasive flow meters mounted to main water headers.



Predictive Alarms

Sensors track shifts across the farm instantly, flagging early drops in water use or ammonia buildups directly onto your dashboard.

ASSET REGISTER VERIFIED

**Every Physical Capital Component Accounted
For**

Lwandile Engineering Projects (Pty) Ltd

Secunda & Kinross Regional Development Hub | Just Energy Transition Submission

PRESTIGE COSMIC VENTURE